

Nebojsa Jovanovic

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Professional Summary

Mechanical Engineer and Software Developer with experience spanning large-scale industrial project execution, commissioning, package engineering, automation, real-time systems, and applied AI. Hatch Engineering Entrance Scholarship recipient and former Intermediate Mechanical Engineer at Hatch. Experienced working both on-site and in engineering/software development environments, with hands-on experience in commissioning, systems handover, industrial automation, edge computing, computer vision, and full-stack development. Seeking opportunities involving site execution, commissioning, industrial automation, controls, or applied software engineering.

Technical Skills

- **Engineering & Systems:** Systems Design, Automation, Real-Time Data Systems, Control Systems, Commissioning, Package Engineering, Edge Computing
- **Languages:** C, C++, Python, Go, JavaScript, TypeScript, C#, Shell, SQL
- **Frameworks & Libraries:** Flask, Django, Aiohttp, Angular, PyTorch, CUDA
- **Tools & Technologies:** Linux, Docker, Podman, Kubernetes (K8s), Neovim (Nvim), Git, Azure DevOps
- **Databases:** PostgreSQL, MongoDB, SQLite
- **Networking:** HTTP, TCP/IP
- **Cloud & MLOps:** Azure ML, Azure DevOps
- **Core Competencies:** Full-Stack Engineering, Machine Learning Integration, Control Systems, Industrial Automation, Edge Devices, Real-Time Systems

Professional Experience

Hatch Ltd. – Mechanical Engineer / Software Developer / Commissioning engineer

Toronto, ON | Jan 2020 – Jul 2026

- Supported commissioning and project execution for the large-scale Jansen potash project, working within the HBJV commissioning team as a Handover Coordinator.
- Coordinated system handover activities and managed documentation and deliverables required to transition systems into commissioning and operations.
- Prepared Handover Packages, Subsystem Test Packages, and Notices of Energization (NOEs) while working directly with project and commissioning teams.
- Optimized project documentation workflows, including automating the generation of NOE documentation and other repetitive document-management tasks.
- Leveraged SmartPlant Foundation, Smart Completions, and SharePoint to manage project documentation, enable contractor execution, and improve commissioning workflows.
- Co-developed VISTA, a real-time hydro energy plant operating system written in C++ and C#, integrating control logic with industrial automation systems.
- Implemented modules for data ingestion and performance analysis while maintaining and modernizing legacy C++ engine and C# UI components.
- Migrated legacy C++ components from an older container library to the modern C++ standard library.
- Worked as part of an Azure DevOps-driven development team supporting software development, testing, and deployment.
- Delivered a computer vision platform for the Onaping Depth Mine, using Python, Flask, PyTorch, and Angular to support safety and operational-efficiency applications.
- Developed and deployed edge computing systems for field-level monitoring and execution of machine-learning models in industrial environments.
- Collaborated with multidisciplinary engineering teams to modernize legacy tools, automate workflows, and improve industrial data-processing processes.
- Applied Azure ML and Azure DevOps to support machine-learning experimentation, deployment, and monitoring.
- Performed package engineering activities across Lithium and Copper processing projects, including Lilac, Li-Cycle, and Manjar Maju Refinery.

- Contributed to tunnel fire ventilation design for the La Fontayne Tunnel and other projects.
 - Developed and tested internal software tools, including a custom web framework and in-memory database for R&D and systems-development work.
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SELECTED PROJECTS

Jansen Commissioning — Handover Coordination

Hatch / HBJV

- Worked on-site as a Handover Coordinator within the HBJV Commissioning team.
- Prepared Handover Packages, Subsystem Test Packages, and Notices of Energization.
- Optimized document-management workflows by developing automation for repetitive documentation tasks, including automatic NOE document generation.
- Used SmartPlant Foundation, Smart Completions, and SharePoint to manage project information, support contractors, and expedite commissioning execution.
- Coordinated information and deliverables across project, engineering, construction, and commissioning stakeholders.

VISTA - Energy Plant Optimization Tool

Hatch

- Contributed to a real-time hydro energy plant operating and optimization system built with C++ and C#.
- Implemented modules for data ingestion and performance analysis.
- Maintained and extended a legacy C++ engine and C# user-interface components.
- Modernized the C++ codebase by migrating functionality from an older container library to the modern C++ standard library.
- Worked within an Azure DevOps-driven software development environment.

VISP - Computer Vision Platform on Edge Devices

Personal Project

- Designed and developed a fully on-premises, scalable infrastructure for deploying YOLO-based computer vision applications on edge devices.
- Built support for adding and managing RTSP cameras and querying stored detections.
- Designed the platform to allow teams to build higher-level, business-logic-driven applications such as alerts, analytics, and operational dashboards.
- Combined computer vision, edge computing, networking, databases, and web services into a single deployable system.

Onaping - Computer Vision in Mining Project

Hatch

- Developed and deployed camera-based inspection and analysis tools using YOLO-based computer vision models.
- Applied machine learning and computer vision techniques to industrial mining applications.

Custom C++ Framework

Personal Project

- Built a lightweight HTTP server and in-memory database engine in C++ to develop a deeper understanding of systems-level architecture.
- Explored networking, data structures, memory management, and software architecture through implementation from first principles.

LiTui — Terminal Chess

Personal Project

- Developed a terminal-based chess application using Go and a text-based user interface.
- Integrated the application with online chess services such as Lichess.
- Built the project as a hands-on exploration of Go, terminal interfaces, networking, and API integration.

Resume Automation

Personal Project

- Developed an automated script that converts this resume from Markdown into a professionally formatted PDF, creating a reproducible document-generation workflow.
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Education

University of Toronto — Bachelor of Applied Science in Mechanical Engineering

Minor in Robotics and Mechatronics | Toronto, ON | 2020

- Hatch Engineering Entrance Scholarship — Full Ride Scholarship Recipient

Continuous Education

- **Harvard CS50** — Completed CS50x, CS50 Python, CS50 Web, and CS50 SQL.
- **edX** — Data Structures and Algorithms — Completed online programming coursework focused on algorithms and data structures.
- **Boot.dev** — Programming Curriculum — Completed gamified programming coursework featuring guided and open-ended projects.
- **Nand2Tetris** — Completed coursework covering computer architecture from NAND gates through a working computer, machine language, assembly, compiler/language development, and operating-system construction, culminating in application development.—

Interests

Large-Scale Project Execution, Site Work, Commissioning, Applied AI in Industry, Control Systems, Energy Technologies, Industrial Automation, Embedded Software, and Systems Engineering.